MeshEcho-CPR: Cost-Responsive Recovery with ACK-Terminated Relay Suppression for LoRa Mesh Networks

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Abstract—Packet loss in a LoRa mesh creates a choice between repeating a known path and injecting another network-wide flood. Repeating too early wastes a stale route, whereas flooding too freely multiplies airtime and collisions. We present MeshEcho-CPR, a source and relay policy that makes this choice at physical transmission time. A routed packet is retried once on the same confirmed path; only after that retry has physically started can one deadline-feasible recovery flood be admitted. A relay records a pending flood and cancels it only after a physically decoded ACK passes flow, request, endpoint, reverse-path, repair-marker, and sender checks, and only before the relay commits the transmission. This preserves a charged wire cost while suppressing redundant relay floods. In a 20-seed paired evaluation with 20 nodes and 6-dB block fading, CPR reaches 0.9829 deadline ACK PDR and 0.9984 destination PDR with 6.29 s mean TX airtime. Under the same paired seed identity, CPR also exceeds legacy MeshEcho, the Meshtastic-like model, and the MeshCore-like model in deadline ACK PDR by 0.0351, 0.0424, and 0.0501, while using 3.94 s, 22.97 s, and 3.61 s less TX airtime per seed, respectively. Results are packet-level simulation evidence for this workload, not measurements of deployed Meshtastic or MeshCore hardware.

Index Terms—LoRa mesh, opportunistic recovery, ACK feedback, flooding suppression, deadline-aware routing, packet-level simulation

I. Introduction

LoRa is attractive for low-power and infrastructure-sparse deployments because it provides a long link budget with inexpensive radios, but its low data rate makes redundant transmissions costly [1]. Multihop LoRa systems therefore combine cached routes, flooding, and local forwarding decisions, as reflected by the two behavior-level product models below. The hard case is a route that was recently confirmed but is no longer reliable: an immediate flood can recover delivery, yet every relay that transmits the same recovery packet consumes airtime and increases collision exposure.

Existing route metrics such as ETX quantify expected transmissions [2], but a metric alone does not specify when a recovery is safe to start or how a relay should react to a terminal ACK that is already on the air. This paper asks whether a protocol can use the physical ordering of repeat, recovery, ACK, and relay-commit events to improve reliability without paying for an unrestricted second flood.

We make three contributions:

- We define MeshEcho-CPR, a bounded source/recovery state machine that separates an initial confirmed-route attempt, one same-path repeat, and at most one deadline-gated recovery flood.
- We add an ACK-terminated relay ledger. Cancellation is accepted only for a physically matching ACK and an uncommitted pending relay, making the suppression event auditable rather than an oracle-based delivery shortcut.
- We provide a seed-paired same-condition comparison with legacy MeshEcho, Meshtastic-like flooding, and MeshCore-like source routing. All rows share topology, traffic, radio configuration, denominators, and seed identities; named products remain separate from simulator-model evidence.

The contribution is a rewritten source, recovery, and feedback core on a preserved packet-level LoRa substrate. The substrate supplies propagation, collision, forwarding, ACK transport, ToA, energy accounting, and evidence ledgers so that the protocols are compared under the same physical model. We do not claim a firmware implementation or universal superiority over every mesh protocol.

II. Related Work

Simulation and measurement studies establish LoRa link and collision models [3]. Route state reduces redundant transmissions when valid, while flooding tolerates stale state at a higher channel cost; CPR focuses on this transition rather than a static score.

The public Meshtastic algorithm description uses delayed managed flooding and duplicate suppression; its newer direct-message path learning is outside the managed-flood arm used here [4]. MeshCore documents a path-oriented unicast mechanism that can learn a path from an initial flood and delivery report, whereas the present MeshCore-like arm isolates the explicit RREQ/RREP/cache exchange [5]. We therefore compare behavior-level models of these mechanisms under one packet-level PHY; the comparison is not a firmware clone or a hardware benchmark.

In the remainder, “Meshtastic-like” means the delayed managed-flood action (z_v) and duplicate-cache timing in (4)– (5); “MeshCore-like” means the cache-hit/discovery

mixture (χ, p_{disc}) in (6)– (8). This naming keeps the comparison reproducible: the equations expose the forwarding decisions that are held constant across seeds, while the footnotes and results do not present them as measurements of the corresponding products.

The closest controls are a deadline-reserved source controller and its no-rescue form. They share the packet model and ACK transport but omit CPR's repeat-before-recovery ordering and ACK-terminated ledger, isolating recovery and relay-cancellation effects.

III. Four-Protocol Models: Formula-Level Meshtastic/MeshCore Comparison

All four arms share one packet-level PHY but make different decisions at the same packet/deadline boundary. We therefore compare behavior-level Meshtastic-like flooding and MeshCore-like source routing with legacy MeshEcho and CPR, not the named products' firmware. Let $P = (v_0, \dots, v_H)$ be a loop-free path, T_D, T_A, and T_F the wire airtime of DATA, ACK, and FLOOD packets, and γ_{SF} the demodulation threshold. Let D_i be the absolute deadline of flow i (we write D when the flow index is clear). For hop h, the simulator's signal model can be written as

$$\begin{aligned} m_h^\ell(t) &= \mathrm{SNR}_h^\ell(t) - \gamma_{SF}, \\ q_h^\ell(t) &= \sigma\big(\kappa m_h^\ell(t)\big), \quad \ell \in \{D, A\}, \\ \sigma(x) &= \frac{1}{1+e^{-x}}. \end{aligned} \tag{1}$$

where q_h^D and q_h^A are the directional no-collision PRR abstractions for DATA and ACK, respectively, and κ is the configured SNR-to-PRR slope. When capture/collision state is present, the packet-level reception draw uses SINR in place of SNR and records the collision separately. For a fixed route, the corresponding DATA and ACK success approximations are

$$\begin{aligned} p_D(P) &= \prod_{h=1}^{H} q^D_{v_{h-1}\to v_h}, \\ p_R(P) &= \prod_{h=1}^{H} q^D_{v_{h-1}\to v_h} q^A_{v_h \to v_{h-1}}. \end{aligned} \tag{2}$$

Here $q^\ell_{i\to j}$ denotes the directional value for link (i, j); the ACK therefore uses the reverse hop. These are explanatory path models; all reported PDRs are obtained from the RX/ACK ledger rather than by assuming independent hops.

a) Legacy MeshEcho route scoring.: The legacy MeshEcho selector is represented by the bottleneck confidence

$$\begin{aligned} m_h(t) &= \mathrm{SNR}_h(t) - \gamma_{SF}, \\ c_h(t) &= \big[\sigma(0.7 m_h(t)) - 0.18\,\mathbf{1}\{\text{collision}_h\}\big]_{0.05}^{0.98}, \\ C_{\text{RREQ}}(P,t) &= \Big[\min_h c_h(t) - \lambda_H \mathbf{1}\{H \ge 2\}\Big]_{0.05}^{0.98}, \\ S_{\text{ME}}(P,t) &= \big[C_{\text{RREQ}}(P,t) - \lambda_A a(t)\big]_0^1, \\ P^*_{\text{ME}} &= \arg\max_{P\in\mathcal{P}} \big(C_{\text{RREQ}}(P,t), -H(P)\big). \end{aligned} \tag{3}$$

where $[x]_a^b = \min(b, \max(a, x))$, $a(t)$ is normalized route age, and λ_H, λ_A are the hop and age penalties. The named MeshEcho runner applies one configured λ_H penalty once a candidate has at least two hops; the indicator above matches that implementation. C_{RREQ} is used for candidate ranking, while S_{ME} is a cache-age diagnostic. In the external runner the fallback confidence threshold is zero, so route miss/discovery timeout, not the age penalty itself, drives fallback; neither score reads the final delivery result.

b) Meshtastic-like managed flooding.: For Meshtastic-like managed flooding, let $\mathcal{V}_F$ be the forwarding-capable relays and let $t_v^{\text{dup}} = \infty$ when no duplicate is decoded before the relay deadline. The delayed-forward commit indicator is

$$\begin{aligned} N_{\text{MT}}^{\text{DATA}} &= 1 + \sum_{v\in\mathcal{V}_F} z_v, \\ N_{\text{MT}}^{\text{TX}} &= N_{\text{MT}}^{\text{DATA}} + |\mathcal{T}_{\text{MT}}^{\text{ACK}}|, \\ A_{\text{MT}} &= \sum_{q\in\mathcal{T}_{\text{MT}}} T(q), \\ p_{\text{MT}}^\ell &\approx 1 - \prod_{P\in\mathcal{P}_F} \big(1 - p_\ell(P)\big), \qquad \ell \in \{A, D\}. \end{aligned} \tag{4}$$

Here $\mathcal{P}_F$ denotes the loop-free source–destination paths exposed by the flood, $p_A(P) = p_R(P)$, $p_D(P)$ is the forward delivery probability, and $\mathcal{T}_{\text{MT}}$ is the set of physically started DATA/ACK transmissions. $N_{\text{MT}}^{\text{TX}}$ is the total started-TX count; the product is only an intuition under independent flood paths; correlated interference is handled by the simulator ledger. For Meshtastic-like managed flooding, the delay and duplicate-suppression action are made explicit by

$$\begin{aligned} m_v &= \mathrm{SNR}_v - \mathrm{SNR}_{\text{req}}(SF), \\ \phi(m) &= 0.6 \left[\frac{8-m}{8}\right]_0^1, \\ w_v &= \max(0.05,\, 0.75 + \phi(m_v) + 0.75 U_v - b_v), \\ b_v &= 0.25 \mathbf{1}\{\text{repeater}_v\}, \\ t_v^{\text{commit}} &= \max(t_v^{\text{rx}} + w_v, t_v^{\text{queue}}), \\ z_v &= \mathbf{1}\big\{t_v^{\text{commit}} < \min\big(t_v^{\text{dup}}, D_i\big)\big\}. \end{aligned} \tag{5}$$

Here $U_v \sim \mathrm{Unif}(0,1)$ is the runner's jitter, t_v^{queue} is the actual queue-available physical start, and D_i is the absolute deadline of flow i. A low SNR margin therefore increases the forwarding delay. A duplicate decoded before commitment makes $z_v = 0$; a duplicate decoded after commitment cannot undo a started TX.

c) MeshCore-like source routing.: For MeshCore-like source routing, let $\chi \in \{0, 1\}$ indicate a valid source-route cache hit. Its cache and discovery-selection rules are

$$\begin{aligned} \chi_{s,d}(t) &= \mathbf{1}\{t \ge t_{\text{install}}(P_{s,d}),\ t - t_{\text{install}}(P_{s,d}) < \text{TTL}\}, \\ P^*_{\text{disc}} &= \begin{cases} P_{\text{first}}, & \overset{W_{\text{disc}}=0}{\mathcal{P}_{\text{RREQ}}\neq\varnothing}, \\ \arg\min\limits_{P\in\mathcal{P}_{\text{RREQ}}} (H(P), P), & \overset{W_{\text{disc}}>0}{\mathcal{P}_{\text{RREQ}}\neq\varnothing}, \\ \varnothing, & \mathcal{P}_{\text{RREQ}} = \varnothing, \end{cases} \\ P_{\text{first}} &= \arg\min_{P\in\mathcal{P}_{\text{RREQ}}} t_{\text{arr}}(P). \end{aligned} \tag{6}$$

Thus a cache miss pays discovery before DATA, while a stale cache can fail without the flood redundancy available to Meshtastic-like forwarding. The four-arm runner uses $W_{\text{disc}} = 0$ (first-reply mode); the second branch is retained to show how a positive collection window would select the shortest candidate. Here $\mathcal{P}_{\text{RREQ}}$ is the set of loop-free candidates and t_{arr} is destination arrival time; an empty candidate set sets $d = 0$. Its exchange cost, including discovery on a miss, uses the convention $P^*_{\text{disc}} = \varnothing$, $p_\ell(\varnothing) = 0$, and $A_{\text{RREP}}(\varnothing) = A_{\text{data}}(\varnothing) = 0$ when $\mathcal{P}_{\text{RREQ}} = \varnothing$:

$$\begin{aligned} A_{\text{data}}(P) &= \sum_{q \in \mathcal{T}_{\text{data}}(P)} T(q) + \sum_{q \in \mathcal{T}_{\text{ack}}(P)} T(q), \\ A_{\text{RREQ}} &= \sum_{q \in \mathcal{T}_{\text{RREQ}}} T(q), \\ A_{\text{RREP}}(P) &= \sum_{q \in \mathcal{T}_{\text{RREP}}(P)} T(q), \\ A_{\text{MC}}(\chi, d) &= \chi A_{\text{data}}(P_{\text{hit}}) \\ &\quad + (1-\chi)\big[A_{\text{RREQ}} + d\big(A_{\text{RREP}}(P^*_{\text{disc}}) \\ &\qquad + A_{\text{data}}(P^*_{\text{disc}})\big)\big]. \end{aligned} \tag{7}$$

The sums retain packet-specific wire lengths; under fixed wire lengths they reduce to hop-count products. When no candidate exists, $d = 0$ and miss-path terms are not evaluated; otherwise they are evaluated only when $d = 1$. The cache mixture therefore has

$$\begin{aligned} p^\ell_{\text{MC}} &= \chi p_\ell(P_{\text{hit}}) + (1-\chi) p_{\text{disc}} p_\ell(P^*_{\text{disc}}), \qquad \ell \in \{A, D\}, \\ \bar{A}_{\text{MC}} &= \chi \bar{A}_{\text{data}}(P_{\text{hit}}) + (1-\chi)\bar{A}_{\text{RREQ}} \\ &\quad + (1-\chi) p_{\text{disc}} \left[\bar{A}_{\text{RREP}} + \bar{A}_{\text{data}}(P^*_{\text{disc}})\right]. \end{aligned} \tag{8}$$

where $\bar{A}_{\text{MC}}$ is expected started-TX cost, $\bar{A}_{\text{RREQ}}$ is the conditional mean RREQ cost on a cache miss, and P_{hit} is the cached path, while P^*_{disc} is the path selected after a successful discovery; d is the per-flow discovery-success indicator and $\bar{A}_{\text{RREP}}$ is the conditional mean started-TX cost of a discovery reply on a cache miss. Also, $p_{\text{disc}} = \Pr\{\exists P : \text{RREQ}(P) \wedge \text{RREP}(P)\}$ is route-discovery success; the independent-path approximation is $1 - \prod_P (1 - r^{\text{req}}_P r^{\text{rep}}_P)$, but the runner's first-reply rule uses the event definition. For managed flooding, z_v is the commit indicator of relay v; for source routing, the three transmission sets count physically started packets.

A. Baseline overhead and algorithmic contrast

Meshtastic-like forwarding spends airtime on committed relay copies, whereas MeshCore-like forwarding spends airtime on discovery after a cache miss. For managed flooding, the source copy and every relay that reaches commitment are charged; only a pending duplicate is suppressible. Thus

$$\begin{aligned} \mathbb{E}[N^{\text{DATA}}_{\text{MT}}] &= 1 + \sum_{v \in \mathcal{V}_F} \Pr\{z_v = 1\}, \\ \mathbb{E}[A_{\text{MT}}] &= T^{(s)}_F + \sum_{v \in \mathcal{V}_F} T_F(v) \Pr\{z_v = 1\} \\ &\quad + \mathbb{E}[A_{\text{ACK}}], \\ S_{\text{MT}} &= \sum_{v \in \mathcal{V}_F} T_F(v)\, \pi^0_v, \\ \pi^0_v &= \Pr\big\{t^{\text{dup}}_v < \min(t^{\text{commit}}_v, D_i)\big\}. \end{aligned} \tag{9}$$

The last term is a pre-commit airtime saving, not a delivery credit.

MeshCore-like source routing trades cache reuse against discovery overhead; a stale route receives no flood credit. CPR instead gates one repeat and one deadline-feasible recovery, with relay cancellation after a matching ACK:

$$\begin{aligned} b_{\text{CPR}}(t) &= \mathbf{1}\{r_t = 1\}\mathbf{1}\{\neg \text{ACK}_t\} \\ &\quad \cdot \mathbf{1}\{\text{valid}(P, g)\}\mathbf{1}\{t + B_F \le D\}. \end{aligned} \tag{10}$$

The same p_{disc} event and first-valid-RREP rule are used for both source-routing branches; the ledger, rather than an independence assumption, captures overlapping paths and collisions.

For a direct comparison, let $A_D(P) \equiv A_{\text{data}}(P)$ count only started DATA/ACK packets, χ_{ME} indicate a valid cached route, and A^{ME}_F be the started fallback-flood cost. The common started-TX cost for one flow is

$$\begin{aligned} C_{\text{ME}} &= \chi_{\text{ME}} A_D(P^*_{\text{ME}}) + (1-\chi_{\text{ME}}) A_{\text{RREQ}} \\ &\quad + (1-\chi_{\text{ME}}) d_{\text{ME}} \big(A_{\text{RREP}}(P^*_{\text{ME}}) + A_D(P^*_{\text{ME}})\big) \\ &\quad + (1-\chi_{\text{ME}})(1-d_{\text{ME}}) A^{\text{ME}}_F, \\ C_{\text{MT}} &= A^{(s)}_F + \sum_{v \in \mathcal{V}_F} z_v T_F(v) + A_{\text{ACK,MT}}, \\ C_{\text{MC}} &= \chi A_D(P_{\text{hit}}) + (1-\chi)\big[A_{\text{RREQ}} + d_{\text{MC}} A_{\text{RREP}}(P^*_{\text{disc}}) \\ &\qquad + d_{\text{MC}} A_D(P^*_{\text{disc}})\big], \\ C_{\text{CPR}} &= I_R \Big[A^{(0)}_R + r A^{(1)}_R + f A_{F,\text{rec}}\Big] \\ &\quad + I_F A_{F,0}. \end{aligned} \tag{11}$$

Here $A^{(s)}_F$ is the source flood cost and $A_{\text{ACK,MT}}$ collects managed-flood ACK transmissions. A failed discovery has $d_{\text{MC}} = 0$, so no fictitious DATA exchange is charged; the ledger realizes C_X. In particular, $A_{F,\text{rec}}$ contains only physically started recovery packets after relay cancellation; a queued but cancelled copy is represented by the separate saving term below, not by a zero-cost delivery shortcut.

To make the comparison executable rather than purely descriptive, let $a_D = p_A(P^*_X)$ denote the branch-local selected path, $a_{\text{hit}} = p_A(P_{\text{hit}})$, $a_{\text{disc}} = p_A(P^*_{\text{disc}})$, and a_F be the conditional ACK success of a flood. Let $d_D = p_D(P^*)$, $d_{\text{hit}} = p_D(P_{\text{hit}})$, $d_{\text{disc}} = p_D(P^*_{\text{disc}})$, and d_F be the corresponding conditional destination-delivery probabilities. With d_{ME} and d_{MC} denoting the corresponding route/discovery-success events, the same branch logic

TABLE I
Behavior-level comparison under the shared packet/PHY model.

	Legacy MeshEcho	Meshtastic-like	MeshCore-like	MeshEcho-CPR
State	score/age route	duplicate cache	source-route cache	generation + ACK ledger
Primary action	route/fallback	delayed flood	RREQ/RREP or route	route/repeat/recovery
Event sequence	cache $\to D$; miss $\to$ RREQ,RREP,DATA/FLOOD	$F_v \to z_v F_v \to ACK$	cache $\to D$; miss $\to$ RREQ,RREP,DATA	$R_0 \to R_1 \to F_{\text{rec}}$
Recovery trigger	miss/fallback	duplicate before commit	cache miss	started repeat + deadline
Relay suppression	no ACK ledger	duplicate before commit	route-directed	matching ACK before commit
Dominant cost	route/fallback	relay redundancy	discovery on miss	bounded repeat/flood
Decision symbols	$S_{\text{MR}}, \chi_{\text{MR}}$	z_v, O_{MT}	$\chi, d_{\text{MC}}, O_{\text{MC}}$	$g_R, g_F, M_v, S_{\text{CPR}}$

gives both confirmation and destination outcomes:

$$\begin{aligned}
\widetilde{p}^{A}_{\text{ME}} &= \chi_{\text{ME}} a_D + (1-\chi_{\text{ME}})[d_{\text{ME}} a_D + (1-d_{\text{ME}}) a_F],\\
\widetilde{p}^{A}_{\text{MT}} &= 1 - \prod_{P \in \mathcal{P}_F} \big(1 - p_A(P)\big),\\
\widetilde{p}^{A}_{\text{MC}} &= \chi a_{\text{hit}} + (1-\chi) p_{\text{disc}} a_{\text{disc}},\\
\widetilde{p}^{D}_{\text{ME}} &= \chi_{\text{ME}} d_D + (1-\chi_{\text{ME}})[d_{\text{ME}} d_D + (1-d_{\text{ME}}) d_F],\\
\widetilde{p}^{D}_{\text{MT}} &= 1 - \prod_{P \in \mathcal{P}_F} \big(1 - p_D(P)\big),\\
\widetilde{p}^{D}_{\text{MC}} &= \chi d_{\text{hit}} + (1-\chi) p_{\text{disc}} d_{\text{disc}},\\
p^{A}_{\text{CPR}} &= I_R[a_0 + \bar{a}_0 g_R a_1 + \bar{a}_0 g_R \bar{a}_1 g_F a_2] + I_F a_{F,0}.
\end{aligned} \tag{12}$$

The tildes mark independent-branch, no-censoring approximations; P^* is the path selected by that branch, while deadline and correlation effects are applied by the packet/ACK ledger. Flooding accumulates paths, source routing pays for a miss, and CPR bounds repeat-before-recovery.

At one source/relay boundary, managed flooding commits relays independently, whereas source routing charges discovery only after a cache miss. Let n_F^{ME} be the started legacy fallback-flood count, $D_h = D(P_{\text{hit}})$, $D_d = D(P^*_{\text{disc}})$, $n_h = n_D(P_{\text{hit}})$, $n_Q = n_R$ be the RREQ/RREP control count, and $n_d = n_R + n_D(P^*_{\text{disc}})$.

$$\begin{aligned}
N_{\text{ME}} &= \chi_{\text{ME}} n_D(P^*_{\text{ME}}) + (1-\chi_{\text{ME}})\left[n_Q + d_{\text{ME}} n_d^{\text{ME}} \right.\\
&\qquad \left. + (1-d_{\text{ME}}) n_F^{\text{ME}}\right],\\
\mathsf{next}_{\text{MT}}(v) &= \begin{cases} F_v, & z_v = 1,\\ \varnothing, & z_v = 0, \end{cases}\\
N_{\text{MT}} &= 1 + \sum_v z_v + n_{\text{ACK,MT}},\\
\mathsf{next}_{\text{MC}}(s,d) &= \begin{cases} D_h, & \chi = 1,\\ Q \to R \to D_d, & \chi = 0,\ d_{\text{MC}} = 1,\\ \text{DROP}, & \chi = 0,\ d_{\text{MC}} = 0, \end{cases}\\
N_{\text{MC}} &= \chi n_h + (1-\chi) n_Q + (1-\chi) d_{\text{MC}} n_d.
\end{aligned} \tag{13}$$

Here $n_d^{\text{ME}} = n_R + n_D(P^*_{\text{ME}})$ and $\varnothing$ is a suppressed relay copy; DROP is a discovery failure, not a successful delivery. For the same ledger, the CPR count is

$$\begin{aligned}
N_{\text{CPR}} = I_R \left[n_R^{(0)} + r n_R^{(1)} + f n_{F,\text{rec}}\right]\\
+ I_F n_{F,0}.
\end{aligned} \tag{14}$$

Table I makes the same distinction visible at the algorithm boundary.

For a uniform four-way comparison, define the outcome vector and the CPR-minus-baseline contrast as

$$\begin{aligned}
\mathbf{y}_X &= \big(p_X^A, p_X^D, \rho_X\big), \qquad \rho_X = \frac{A_X}{MT_D},\\
\boldsymbol{\delta}_X &= \big(\delta_X^A, \delta_X^D, \delta_X^\rho\big),\\
\delta_X^A &= p^A_{\text{CPR}} - p_X^A, \qquad \delta_X^D = p^D_{\text{CPR}} - p_X^D,\\
\delta_X^\rho &= \rho_X - \rho_{\text{CPR}},\\
g_X^A &= \frac{\delta_X^A}{p_X^A}, \qquad g_X^D = \frac{\delta_X^D}{p_X^D}, \qquad g_X^\rho = \frac{\delta_X^\rho}{\rho_X},\\
\text{CPR} \succeq X &\iff \min\{\delta_X^A, \delta_X^D, \delta_X^\rho\} \geq 0.
\end{aligned} \tag{15}$$

The first two coordinates reward reliability and the third rewards lower CPR airtime; g_X^A, g_X^D, g_X^ρ are the corresponding relative gains, and ρ_X uses the unmarked reference DATA ToA. The paired table is the ledger realization for $X \in \{\text{ME}, \text{MT}, \text{MC}\}$.

The common cost vector is complemented by three mechanism-specific ledger observables. They isolate the extra work or the saved work without pretending that source, ACK, discovery, and fallback terms are interchangeable:

$$\begin{aligned}
O_{\text{MT}} &= \sum_{v \in \mathcal{V}_F} z_v T_F(v),\\
O_{\text{MC}} &= (1-\chi)\big[A_{\text{RREQ}} + d_{\text{MC}} A_{\text{RREP}}(P^*_{\text{disc}})\\
&\qquad + d_{\text{MC}} A_D(P^*_{\text{disc}})\big],\\
S_{\text{CPR}} &= \sum_{v \in \mathcal{V}_F} T_F(v)\\
&\qquad \cdot \mathbf{1}\{M_v = 1,\ \tau_v^{\text{ack}} < \tau_v^{\text{commit}}\}.
\end{aligned} \tag{16}$$

Here O_{MT} is committed relay airtime, O_{MC} is cache-miss discovery plus the resulting routed exchange, and S_{CPR} is the airtime removed by a matching ACK before relay commitment. The full per-flow costs remain those in (11); these three terms are diagnostics that explain why the same physical ledger produces different protocol totals.

For the reported paired evaluation, the same contrast is written directly as

$$\begin{aligned}
(\Delta^A_{\text{ME}}, \Delta^A_{\text{MT}}, \Delta^A_{\text{MC}}) &= (0.0351,\ 0.0424,\ 0.0501),\\
(\Delta^D_{\text{ME}}, \Delta^D_{\text{MT}}, \Delta^D_{\text{MC}}) &= (0.0368,\ 0.0098,\ 0.0533),\\
(\Gamma_{\text{ME}}, \Gamma_{\text{MT}}, \Gamma_{\text{MC}}) &= (0.385,\ 0.785,\ 0.364),
\end{aligned} \tag{17}$$

where $\Delta_X^\ell = p^\ell_{\text{CPR}} - p_X^\ell$ and $\Gamma_X = 1 - \bar{A}_{\text{CPR}}/\bar{A}_X$ for the frozen workload.

All four families use the same LoRa time-on-air accounting. For payload length L, spreading factor SF, bandwidth BW, coding-rate index CR, and the usual header/CRC flags, the packet cost is

$$\begin{aligned}
T(q) &= (N_{\text{pre}} + 4.25 + N_{\text{pl}}) \frac{2^{SF}}{BW},\\
K_{\text{pl}} &= 8L - 4SF + 28 + 16CRC - 20IH,\\
N_{\text{pl}} &= 8 + \max\left(\left\lceil \frac{K_{\text{pl}}}{4(SF - 2DE)} \right\rceil (CR+4), 0\right).
\end{aligned} \tag{18}$$

Here $L = L(q)$ is the PHY payload length, $N_{\text{pre}} = 8$, $CRC = 1$, $IH = 0$ denotes explicit headers, DE is low-data-rate optimization, and $CR = 1$ denotes coding rate 4/5. This shared ToA expression is why the comparison reports physical started TX airtime rather than a protocol-specific packet count.

For every arm X, the reported physical cost is computed from the same TX ledger rather than from a protocol-

specific packet estimate:

$$
\begin{aligned}
A_X &= \sum_{q\in\mathcal{T}_X} T(q), \qquad E_X = VI_{\mathrm{TX}}A_X,\\
\widehat{p}_X^{A} &= \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}\{t_{X,i}^{\mathrm{ACK}} \le D_i\},\\
B_X &= 8\sum_{i=1}^{M} L_i\mathbf{1}\{t_{X,i}^{\mathrm{DEL}} \le D_i\},\\
\widehat{p}_X^{D} &= \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}\{t_{X,i}^{\mathrm{DEL}} \le D_i\},\\
\eta_X^{\mathrm{air}} &= \frac{B_X}{A_X}, \qquad \eta_X^{\mathrm{energy}} = \frac{B_X}{E_X}, \qquad \rho_X = \frac{A_X}{MT_D}.
\end{aligned}
\tag{19}
$$

Here $\mathcal{T}_X$ contains physically started packets, VI_{TX} is modeled TX power, and M is the common flow denominator; T_D is the reference DATA ToA. A lost reverse ACK can therefore lower ACK PDR without changing destination PDR.

For seed k, let $Y_{X,k}$ be any one of the reported seed-level metrics. The paired comparison is defined by

$$
\begin{aligned}
\Delta_{X,k} &= Y_{\mathrm{CPR},k} - Y_{X,k},\\
\mathrm{CI}_{95}(\Delta_X) &= \bar{\Delta}_X \pm t_{0.975,19}\frac{s_{\Delta_X}}{\sqrt{20}}.
\end{aligned}
\tag{20}
$$

where $\bar{\Delta}_X$ and s_{Δ_X} are the mean and sample standard deviation over the same 20 seeds. This keeps the comparison paired by topology, application trace, radio configuration, and pair pool.

IV. MeshEcho-CPR

A. Local state and action ordering

Each unicast flow has a source, destination, enqueue time, original deadline, and a bounded route record. A route is usable only when its endpoint sequence, generation, expiry, and local state are valid. At the actual queue-available TX start, the source computes conservative upper bounds B_R and B_F for a routed DATA/ACK exchange and a recovery FLOOD/ACK exchange, including packet ToA, forwarding delay, and a safety margin. The source does not inspect a future reception outcome or a global PRR oracle.

The initial action is R_{initial} when a confirmed route fits before the original deadline; otherwise it is F_{initial} when a flood fits. After an unacknowledged routed initial transmission, CPR schedules one same-path repeat. The repeat request is not considered a transmission until the packet physically starts. Let g denote the route generation paired with path P. Only then can the recovery decision admit one flood, and only if the complete bound still fits:

$$
\begin{aligned}
&R_{\mathrm{initial}} \to R_{\mathrm{repeat}},\\
&R_{\mathrm{repeat}} \to F_{\mathrm{recovery}} \quad \text{iff } r_t \wedge \neg\mathrm{ACK}_t,\\
&\qquad\qquad \mathrm{valid}(P,g) \wedge t + B_F \le t_{\mathrm{deadline}}.
\end{aligned}
\tag{21}
$$

If the flow is already acknowledged, the route generation is stale, or the bound does not fit, no recovery request is started. This ordering prevents a queued but never-started repeat from authorizing a flood.

Let $I_R, I_F \in \{0,1\}$ denote the initial routed and initial flood branches. For an admitted flow $I_R + I_F = 1$; a deadline drop has $I_R + I_F = 0$. The local admission predicate for action a is

$$
\begin{aligned}
g_a(t,D) &= \mathbf{1}\{t + B_a \le D\},\\
a^*(t) &= \arg\max_{a\in\{R,F\}} g_a(t,D)\,\mathrm{valid}(a),
\end{aligned}
\tag{22}
$$

where $\mathrm{valid}(a)$ is the local route/flood availability indicator; if both products are zero, $a^*(t) = \mathrm{DROP}$. A deterministic tie rule favors the confirmed route. The predicate is evaluated at physical TX start, not when a request is merely queued.

For a path with H hops, CPR computes the route budget from the actual wire exchange rather than from a delivery prediction:

$$
\begin{aligned}
B_R(P) &= \sum_{h=1}^{H}\left[T_D(h) + T_A(h)\right] + \mu_R,\\
B_F(H_{\max}) &= \sum_{h=1}^{H_{\max}}\left[T_F(h) + T_A^F(h) + \delta_F(h)\right] + \mu_F.
\end{aligned}
\tag{23}
$$

where $H_{\max}$ is the configured hop bound and $\delta_F(h)$ is the worst-case flood forwarding/backoff delay, with $\delta_F(1) = 0$ for the source transmission and forwarding delay reserved only for later hops; μ_R, μ_F are safety margins. The route bound matches the simulator's ToA-plus-margin admission; the flood bound additionally reserves forwarding and backoff delay. Marker-dependent packet lengths are retained in T_D, T_A, T_F, T_A^F. Let a_0, a_1, a_2 be conditional ACK successes for the initial route, repeat, and recovery flood after preceding ACK failures; let d_0, d_1, d_2 be conditional destination deliveries, with d_1 conditioned on initial ACK failure and destination failure and d_2 additionally on repeat ACK failure and destination failure. Let $a_{F,0}$ and $d_{F,0}$ denote initial-flood outcomes. With $\bar{x} = 1 - x$, let t_R be the queue-available start time of the repeat and t_F the current recovery-admission time. The guards are

$$
\begin{aligned}
g_R &= \mathbf{1}\{r_R \wedge \mathrm{valid}(P,g) \wedge t_R + B_R \le D\},\\
g_F &= \mathbf{1}\{g_R \wedge \mathrm{valid}(P,g,t_F) \wedge \neg\mathrm{ACK}_{t_F} \wedge t_F + B_F \le D\}.
\end{aligned}
\tag{24}
$$

Each predicate is evaluated at that action's physical queue-available TX time,

$$
p_{\mathrm{CPR}}^{A} = I_R[a_0 + \bar{a}_0 g_R a_1 + \bar{a}_0 g_R \bar{a}_1 g_F a_2] + I_F a_{F,0}. \tag{25}
$$

Here r_R means that the repeat packet physically started; g_R therefore guards both repeat terms, while g_F additionally guards the recovery budget from the later time t_F. The already-spent time is therefore not reset when the recovery branch is evaluated. The conditional quantities a_1, d_1 are measured only after $g_R = 1$. The destination-delivery probability does not use destination failure to trigger a retry: a DATA packet may arrive while its reverse ACK fails. Therefore

$$
\begin{aligned}
p_{\mathrm{CPR}}^{D} = {} & I_R d_0 + I_R \bar{a}_0 g_R \bar{d}_0 d_1\\
& + I_R \bar{a}_0 g_R \bar{d}_0 \bar{a}_1 \bar{d}_1 g_F d_2 + I_F d_{F,0}.
\end{aligned}
\tag{26}
$$

These equations make both initial branches and the one-repeat/one-recovery limits explicit while keeping ACK PDR and destination PDR as separate observables.

The same ordering gives a per-flow airtime decomposition. With r and f indicating a physically started repeat and recovery flood, and z_v marking a committed recovery relay, CPR pays

$$\begin{aligned} A_R^{+}(P) &= \sum_{h=1}^{H} \left[T_D(h) + T_A(h)\right], \\ A_R^{(j)}(P) &= \sum_{q \in \mathcal{T}_R^{(j)}(P)} T(q), \quad j \in \{0,1\}, \\ A_{F,j} &= T_F^{(s)} + T_A^{F,j} + \sum_{v \in \mathcal{V}_{F,j}} z_{v,j} T_F(v), \\ A_{\mathrm{CPR}}(P) &= I_R \left[A_R^{(0)} + r A_R^{(1)} + f A_{F,\mathrm{rec}}\right] + I_F A_{F,0}. \end{aligned} \tag{27}$$

The full-exchange quantity A_R^{+} is a planned bound; all other terms are sums over physically started packets. The repeat marker may change wire length, while a queued but cancelled relay is absent from $A_{F,\mathrm{rec}}$. Forwarding delay and safety margins affect admission, not this ToA sum.

B. ACK identity and route generation

The source accepts an ACK only when its identifiers, endpoints, reverse path, sender, deadline, marker, and physical-start record match the current flow; the marker selects initial, repeat, or recovery. A valid ACK commits the route generation, while a previous-generation ACK is inert.

$$\begin{aligned} \mathcal{I}_{\mathrm{ACK}} &= I_{\mathrm{id}} I_{\mathrm{endpoint}} I_{\mathrm{path}} I_{\mathrm{sender}} \\ &\quad \cdot I_{\mathrm{marker}} I_{\mathrm{start}} I_{\mathrm{deadline}} I_{\mathrm{generation}}, \\ \mathrm{acceptACK} &= \mathcal{I}_{\mathrm{ACK}}. \end{aligned} \tag{28}$$

Here each $I_c = \mathbf{1}\{c \text{ matches the flow record}\}$; a failed parser/state factor leaves the flow and relay ledger unchanged.

C. ACK-terminated relay suppression

When a relay decodes a CPR flood, it creates a bounded handle keyed by relay, origin, flow, request, and marker, with endpoint and forward-path context. It cancels the queued request only when

$$\begin{aligned} \mathsf{cancel} &= \mathsf{match}(\text{flow}, \text{request}, \text{endpoints}, \text{marker}) \\ &\quad \wedge \mathsf{valid}(\text{reverse path}, \text{path index}, \text{sender}) \\ &\quad \wedge \neg \mathsf{committed}. \end{aligned} \tag{29}$$

The ACK path must be loop-free, endpoint-consistent, and on the reverse hop; mismatched fields or committed handles are rejected. Source generation checks remain separate, and only started TX is charged.

Let τ_v^{ack} and τ_v^{commit} be the arrival and commit times for relay v, and let M_v indicate the flow/request, endpoint, marker, reverse-path, and sender match in (29); set $\tau_v^{\mathrm{ack}} = \infty$ when no matching ACK is decoded. The expected relay airtime is

$$\begin{aligned} \mathbb{E}[A_{\mathrm{relay}}] &= \sum_{v \in \mathcal{V}_F} T_F(v) \Pr\left(M_v = 0 \ \vee \ \tau_v^{\mathrm{ack}} \geq \tau_v^{\mathrm{commit}}\right), \\ \Delta A &= \sum_{v \in \mathcal{V}_F} T_F(v) \Pr\left(M_v = 1 \wedge \tau_v^{\mathrm{ack}} < \tau_v^{\mathrm{commit}}\right). \end{aligned} \tag{30}$$

Thus CPR charges every committed relay and saves airtime only when a matching ACK arrives before commit; bounded eviction is logged.

V. Experimental Method

A. Frozen case and cohorts

The frozen case has 20 nodes in a 3000 m square, four

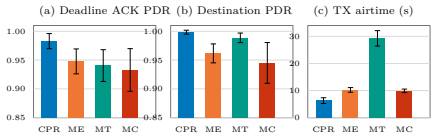


Fig. 1. Four-protocol paired means; error bars are nominal 95% seed intervals.

TABLE II
Paired means over 20 seeds; PDR is deadline-based and energy is modeled TX energy.

Arm	ACK PDR	Dest. PDR	TX airtime (s)	TX energy (J)	TX count	RX attempts
CPR	0.9829	0.9984	6.290	2.491	81.65	1633
Legacy MeshEcho†	0.9477	0.9616	10.231	4.052	156.50	3130
Meshtastic-like†	0.9405	0.9886	29.257	11.586	318.25	6365
MeshCore-like†	0.9328	0.9451	9.896	3.919	153.10	3062

deadline over 240 s plus a 30-s drain. The four arms use the same scheduled traces for seeds 92300–92319; route TTL is 600 s, MeshCore-like discovery uses first-reply mode ($W_{\mathrm{disc}} = 0$), and managed-flood delay uses base 0.45 s plus jitter $0.65U_v$. TX energy uses 3.3 V and 120 mA and excludes RX. This is a behavior-level paired evaluation, not an untouched holdout claim or a hardware benchmark.

VI. Results

A. Paired performance

Table II gives the direct four-protocol comparison; the dagger rows are behavior-level models rather than product measurements.

To expose the mechanism difference, let H_k, M_k be route-cache hits and misses, R_k, A_k successful and attempted discoveries, S_k suppressed forwards, D_k duplicate receptions, and $C_{X,k}, N_{X,k}$ control and total started transmissions for arm X. The seed-balanced observables are

$$\begin{aligned} h_k &= \frac{H_k}{\max(1, H_k + M_k)}, \qquad p_{\mathrm{disc},k} = \frac{R_k}{\max(1, A_k)}, \\ s_{\mathrm{dup},k} &= \frac{S_k}{\max(1, D_k)}, \qquad o_{X,k} = \frac{C_{X,k}}{\max(1, N_{X,k})}, \\ \bar{q} &= K^{-1} \sum_{k=1}^{K} q_k, \quad q \in \{h, p_{\mathrm{disc}}, s_{\mathrm{dup}}, o_X\}. \end{aligned} \tag{31}$$

Here s_{dup} is a normalized counter; zero-denominator seeds contribute zero. Across 20 seeds, $(h_{\mathrm{ME}}, p_{\mathrm{disc,ME}}) = (0.8647, 0.9433)$, $(h_{\mathrm{MC}}, p_{\mathrm{disc,MC}}) = (0.8610, 0.9550)$, and $(s_{\mathrm{dup,MT}}, s_{\mathrm{dup,CPR}}) = (0.5141, 0.1162)$.

For clarity, the four-arm mechanism vector is

$$\begin{aligned} \mathbf{m}_X &= \left(\bar{h}_X, \bar{p}_{\mathrm{disc},X}, \bar{s}_{\mathrm{dup},X}, \bar{o}_X\right), \\ \bar{o}_X &= \frac{1}{K} \sum_{k=1}^{K} \frac{C_{X,k}}{\max(1, N_{X,k})}. \end{aligned} \tag{32}$$

The resulting ledger values are written below; $\perp$ means that the mechanism is not part of that arm, not that its value is zero. This makes the comparison between managed flooding and source routing explicit: Meshtastic-like forwarding exposes duplicate suppression but no route-discovery cache, whereas MeshCore-like forwarding exposes cache hits and RREQ/RREP discovery but no

flood-level suppression. For the coordinate order in (32), the corresponding vectors are

$$\begin{aligned} \mathbf{m}_{\mathrm{ME}} &= (0.8647, 0.9433, 0.0130, 0.7625), \\ \mathbf{m}_{\mathrm{MT}} &= (\perp, \perp, 0.5141, 0.1116), \\ \mathbf{m}_{\mathrm{MC}} &= (0.8610, 0.9550, 0, 0.7741), \\ \mathbf{m}_{\mathrm{CPR}} &= (\perp, \perp, 0.1162, 0.4567). \end{aligned} \tag{33}$$

Here $\perp$ denotes a non-applicable mechanism, not a measured zero.

a) Formula-level baseline comparison.: The four rows differ at the physical-TX boundary. Let D_{hit}, D_{disc}, and D_{route} denote routed DATA/ACK airtime, and let Q_{disc} and Q_{RREQ} denote discovery/control airtime. The same physical ledger then gives

$$\begin{aligned} A_{\mathrm{MT}} &= T_F^{(s)} + \sum_{v \in \mathcal{V}_F} z_v T_F(v) + A_{\mathrm{ACK,MT}}, \\ A_{\mathrm{MC}} &= \chi D_{\mathrm{hit}} + (1-\chi)\big[Q_{\mathrm{disc}} + d_{\mathrm{MC}} D_{\mathrm{disc}}\big], \\ A_{\mathrm{ME}} &= \chi_{\mathrm{ME}} D_{\mathrm{route}} + (1-\chi_{\mathrm{ME}}) Q_{\mathrm{RREQ}} \\ &\quad + (1-\chi_{\mathrm{ME}})\big[d_{\mathrm{ME}} D_{\mathrm{route}} \\ &\quad + (1-d_{\mathrm{ME}}) A_F^{\mathrm{fallback}}\big], \\ A_{\mathrm{CPR}} &= I_R\big[D_0 + r D_1 + f A_{F,\mathrm{rec}}\big] + I_F A_{F,0}. \end{aligned} \tag{34}$$

Here z_v is a committed relay, d_{MC} and d_{ME} are discovery-success indicators, and r, f are physical repeat/recovery starts: MT pays relay copies, MC pays Q_{disc} on a miss, ME may pay fallback, and CPR bounds one repeat before one recovery. For $G_X^A = \bar{A}_X / \bar{A}_{\mathrm{CPR}}$ and $G_X^N = \bar{N}_X / \bar{N}_{\mathrm{CPR}}$, Table II gives $(G_{\mathrm{ME}}^A, G_{\mathrm{MT}}^A, G_{\mathrm{MC}}^A) = (1.63, 4.65, 1.57)$ and $(G_{\mathrm{ME}}^N, G_{\mathrm{MT}}^N, G_{\mathrm{MC}}^N) = (1.92, 3.90, 1.88)$; here $\bar{N}_X$ is the mean started-request count per seed. All rows are behavior-level models.

b) Conclusion.: MeshEcho-CPR bounds repeat-before-recovery and ACK-terminated suppression in packet-level simulation.

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